Snowflake and PowerBI Integration

Create Warehouse:

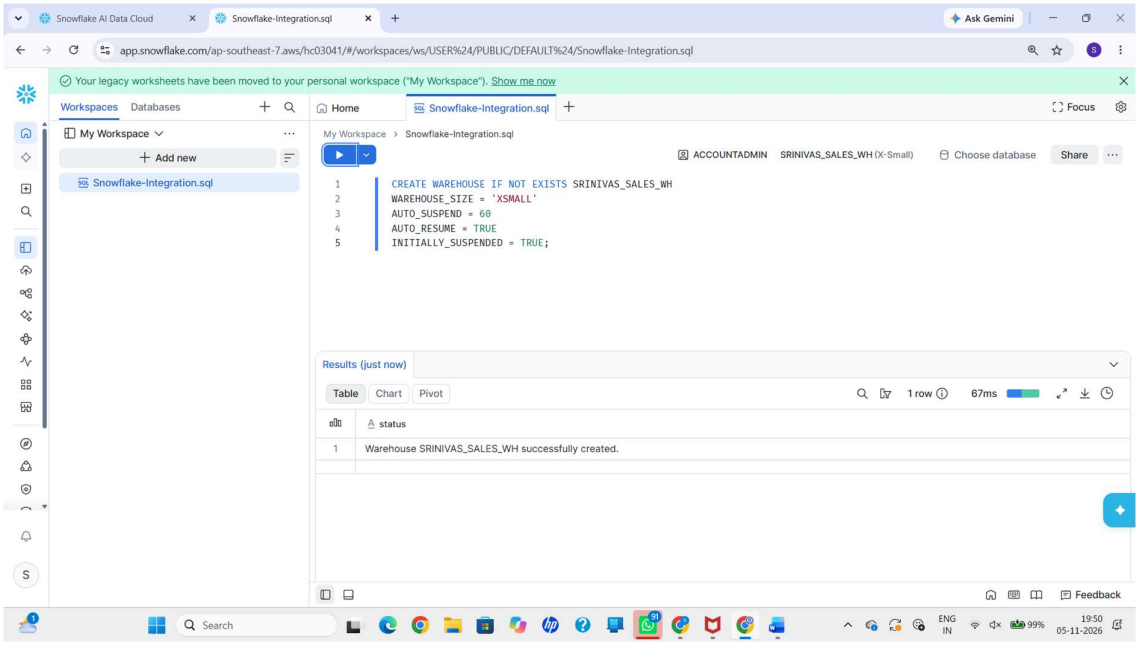
```
CREATE WAREHOUSE IF NOT EXISTS SRINIVAS_SALES_WH

WAREHOUSE_SIZE = 'XSMALL'

AUTO_SUSPEND = 60

AUTO_RESUME = TRUE

INITIALLY_SUSPENDED = TRUE;
```



Create Database and Schema:

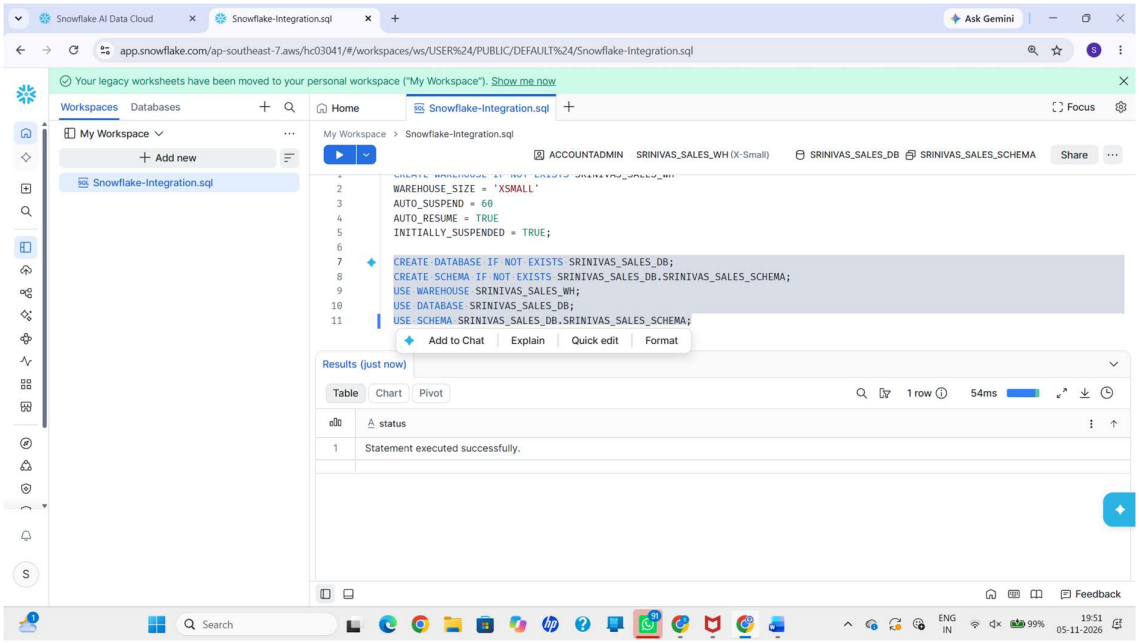
```
CREATE DATABASE IF NOT EXISTS SRINIVAS_SALES_DB;

CREATE SCHEMA IF NOT EXISTS SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA;

USE WAREHOUSE SRINIVAS_SALES_WH;

USE DATABASE SRINIVAS_SALES_DB;

USE SCHEMA SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA;
```



Create File Format and Stage:

```
CREATE OR REPLACE FILE FORMAT
SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES_CSV_FORMAT

TYPE = 'CSV'

SKIP_HEADER = 1

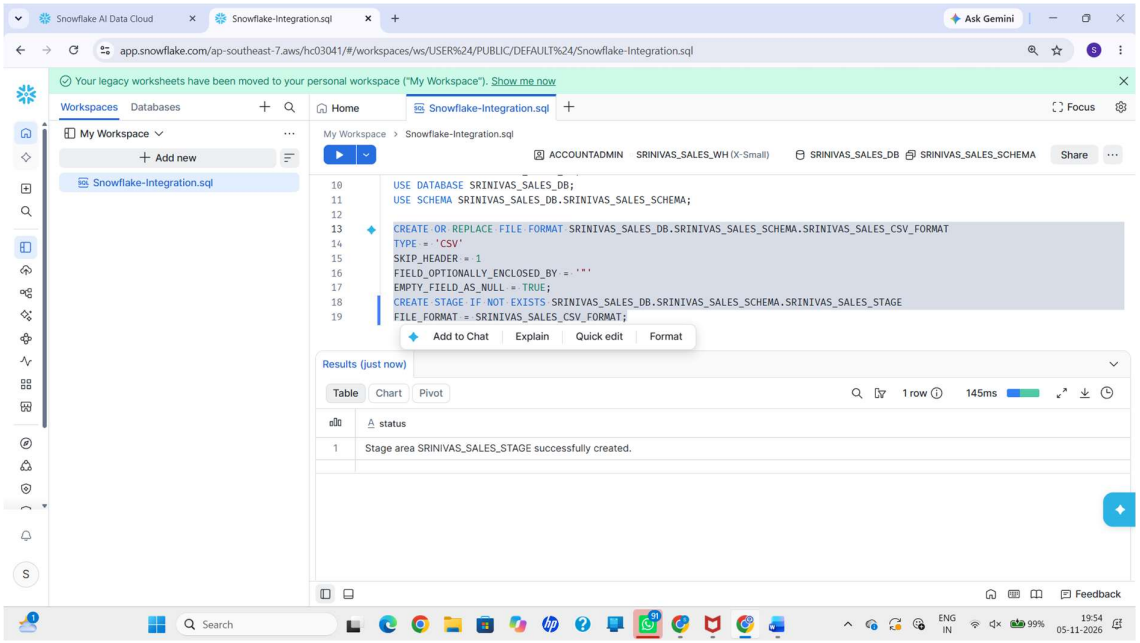
FIELD_OPTIONALLY_ENCLOSED_BY = '"'

EMPTY_FIELD_AS_NULL = TRUE;



CREATE STAGE IF NOT EXISTS SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES_STAGE

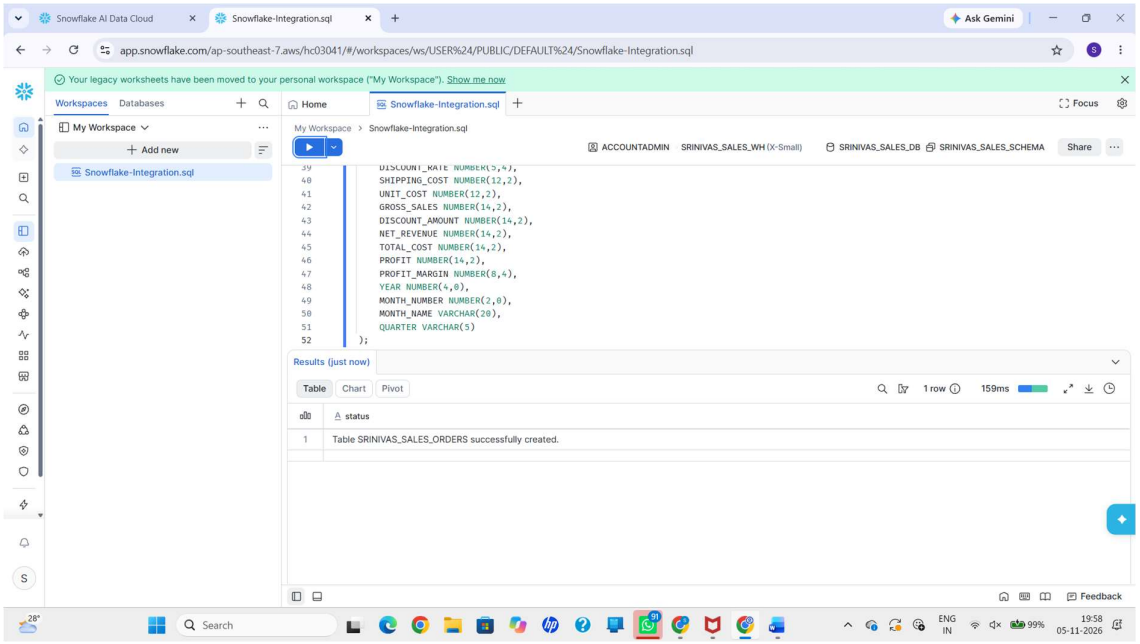
FILE_FORMAT = SRINIVAS_SALES_CSV_FORMAT;
```



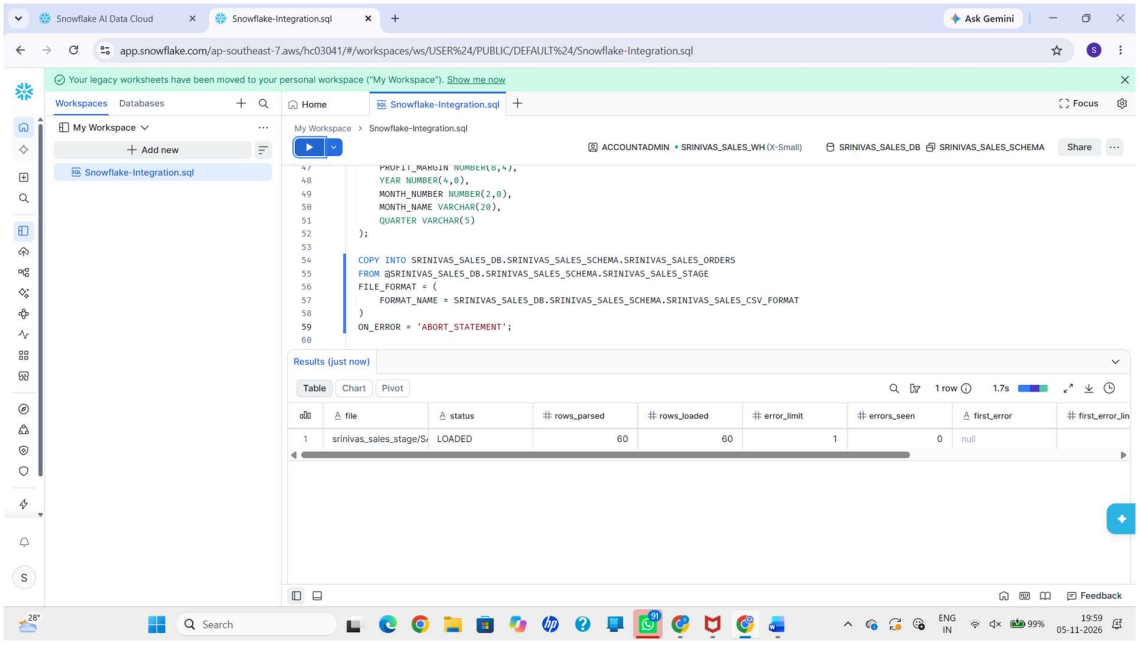
Create Sales Table :

```
CREATE OR REPLACE TABLE SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES_ORDERS
(
    ORDER_ID VARCHAR(20),
    ORDER_DATE DATE,
    CUSTOMER_ID VARCHAR(20),
    CUSTOMER_NAME VARCHAR(100),
    REGION VARCHAR(30),
    STATE VARCHAR(50),
    CITY VARCHAR(50),
    PRODUCT_CATEGORY VARCHAR(50),
    PRODUCT_NAME VARCHAR(100),
    SALES_CHANNEL VARCHAR(30),
    PAYMENT_METHOD VARCHAR(30),
    ORDER_STATUS VARCHAR(30),
    QUANTITY NUMBER(10,0),
    UNIT_PRICE NUMBER(12,2),
    DISCOUNT_RATE NUMBER(5,4),
    SHIPPING_COST NUMBER(12,2),
```

```
  UNIT_COST NUMBER(12,2),
  GROSS_SALES NUMBER(14,2),
  DISCOUNT_AMOUNT NUMBER(14,2),
  NET_REVENUE NUMBER(14,2),
  TOTAL_COST NUMBER(14,2),
  PROFIT NUMBER(14,2),
  PROFIT_MARGIN NUMBER(8,4),
  YEAR NUMBER(4,0),
  MONTH_NUMBER NUMBER(2,0),
  MONTH_NAME VARCHAR(20),
  QUARTER VARCHAR(5)
);
```



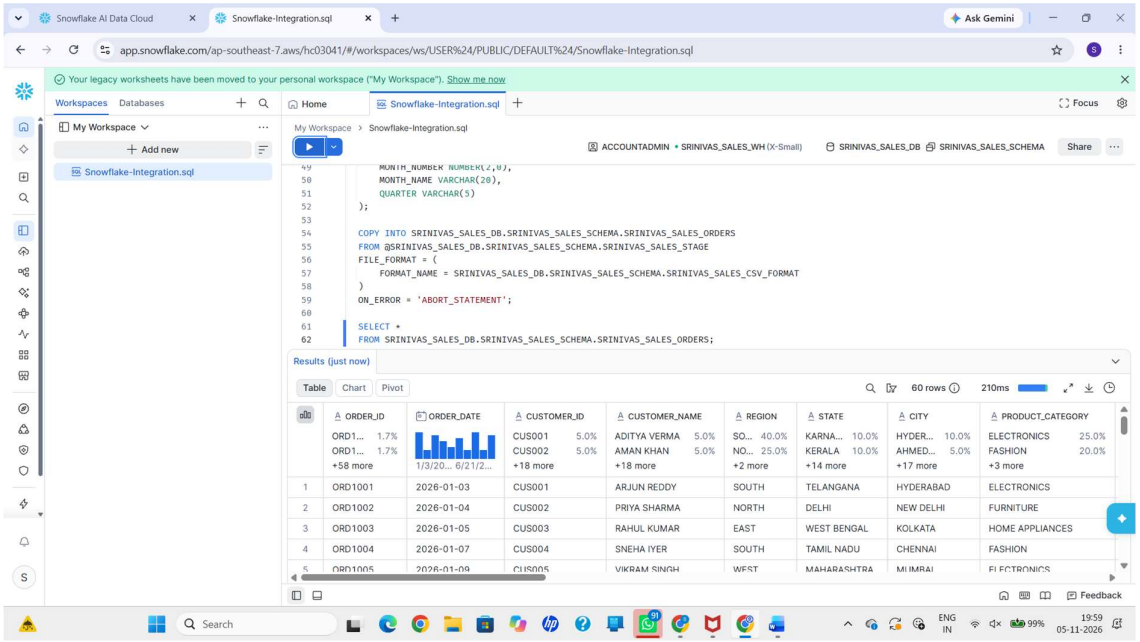

```
COPY INTO SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES_ORDERS
FROM @SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES_STAGE
FILE_FORMAT = (
  FORMAT_NAME = SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES_CSV_FORMAT
)
ON_ERROR = 'ABORT_STATEMENT';
```



Display all records:

```
SELECT *

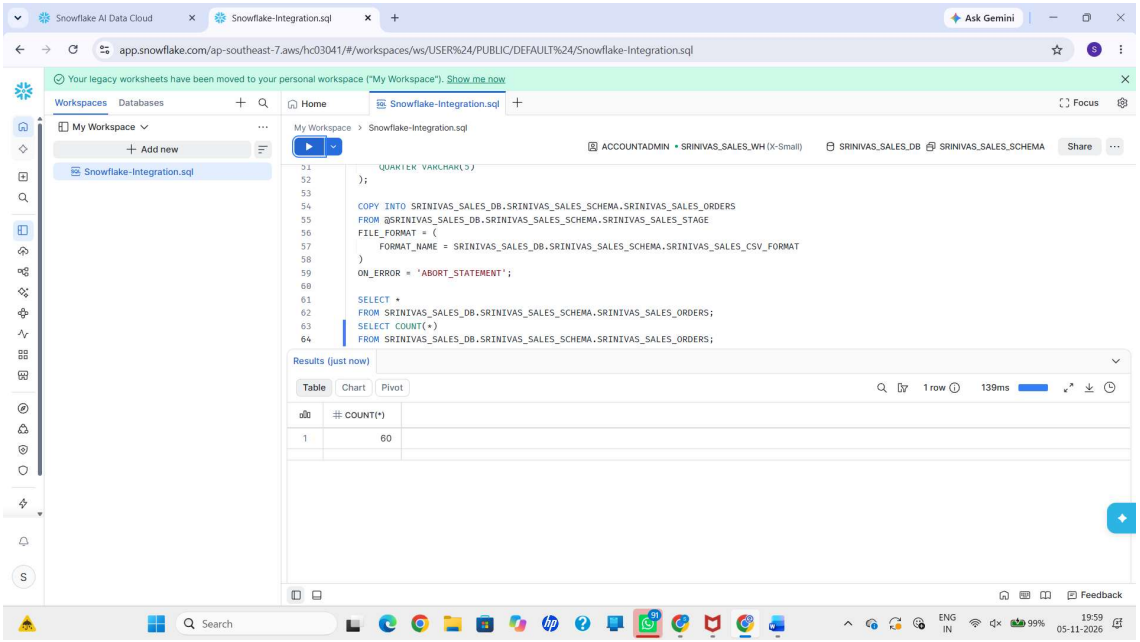
FROM SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES_ORDERS;
```



Check total records:

```
SELECT COUNT(*)

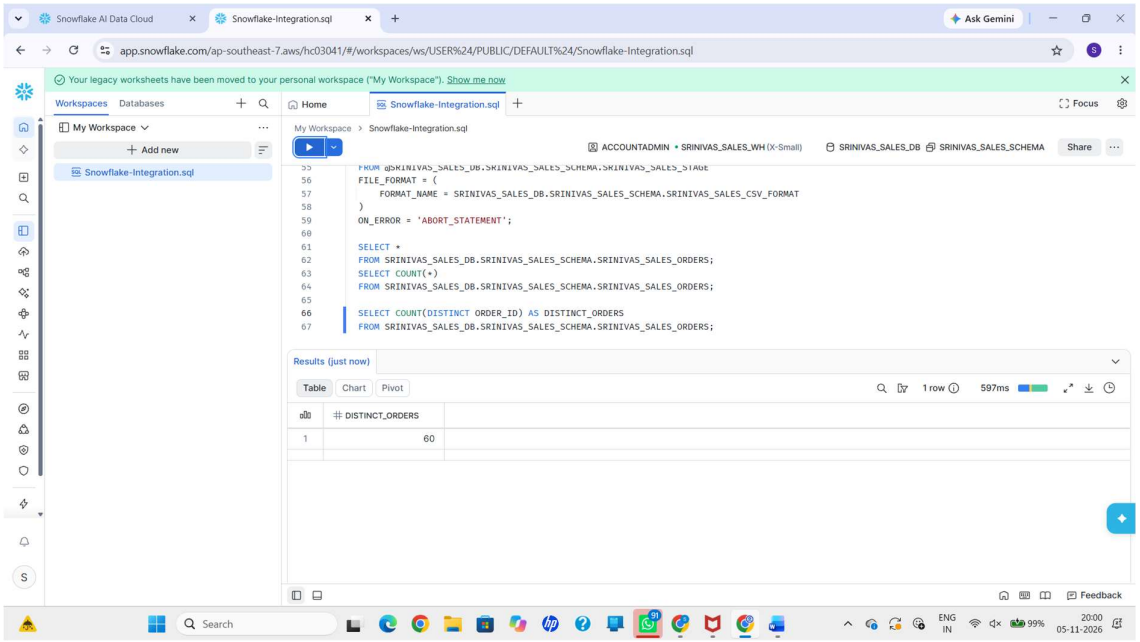
FROM SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES_ORDERS;
```



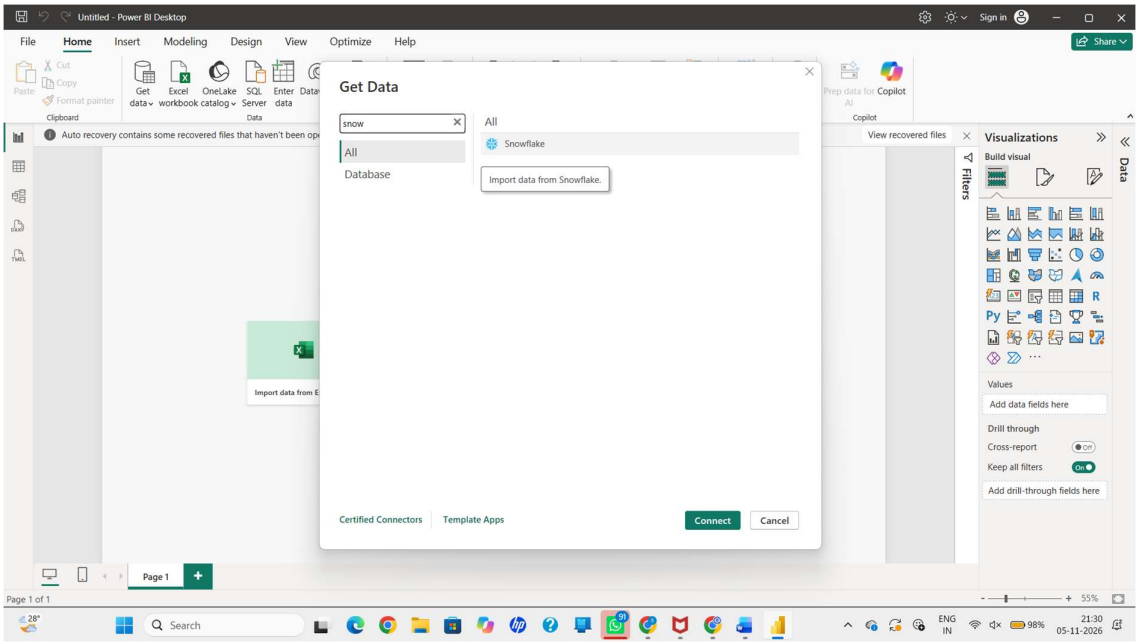
To count unique orders :

```
SELECT COUNT(DISTINCT ORDER_ID) AS DISTINCT_ORDERS

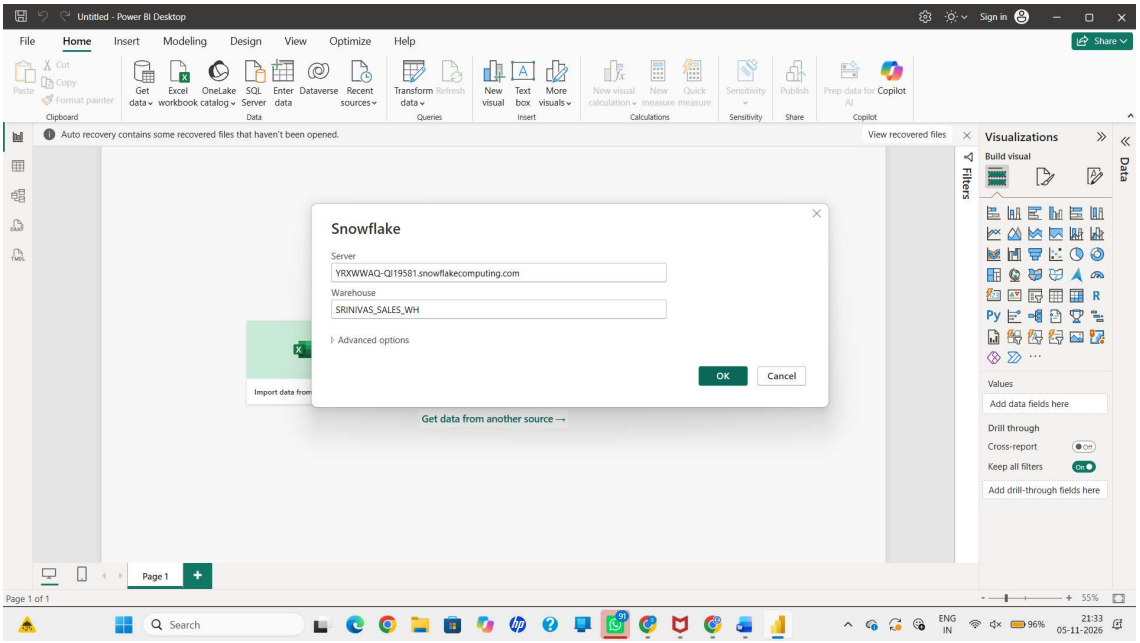
FROM SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES_ORDERS;
```



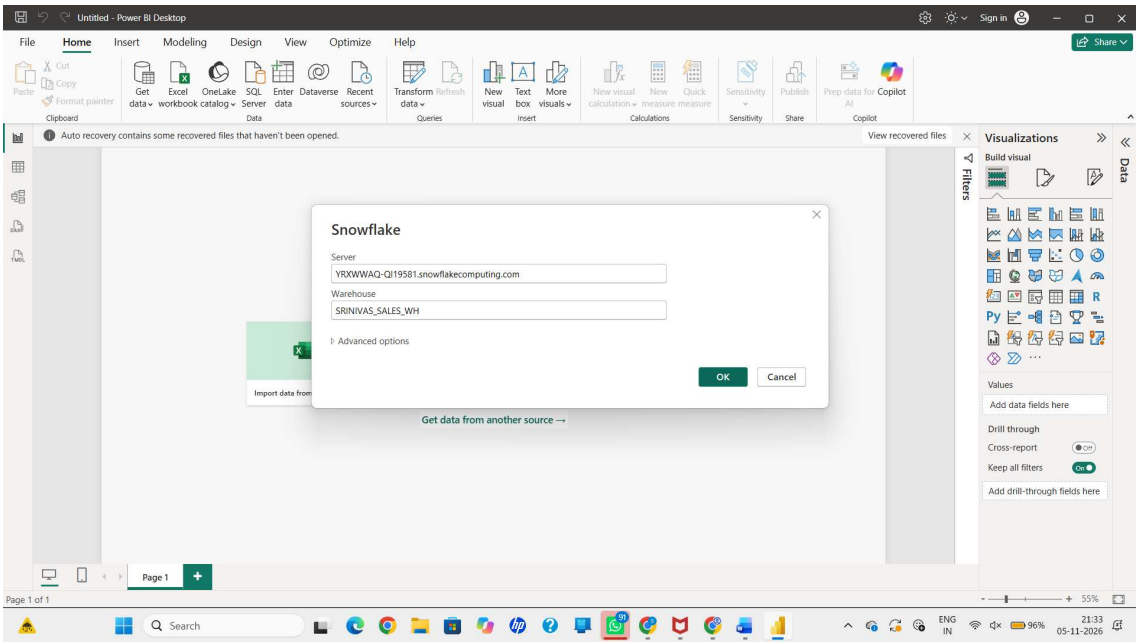
Importing Data from Snowflake using Account/Server URL:



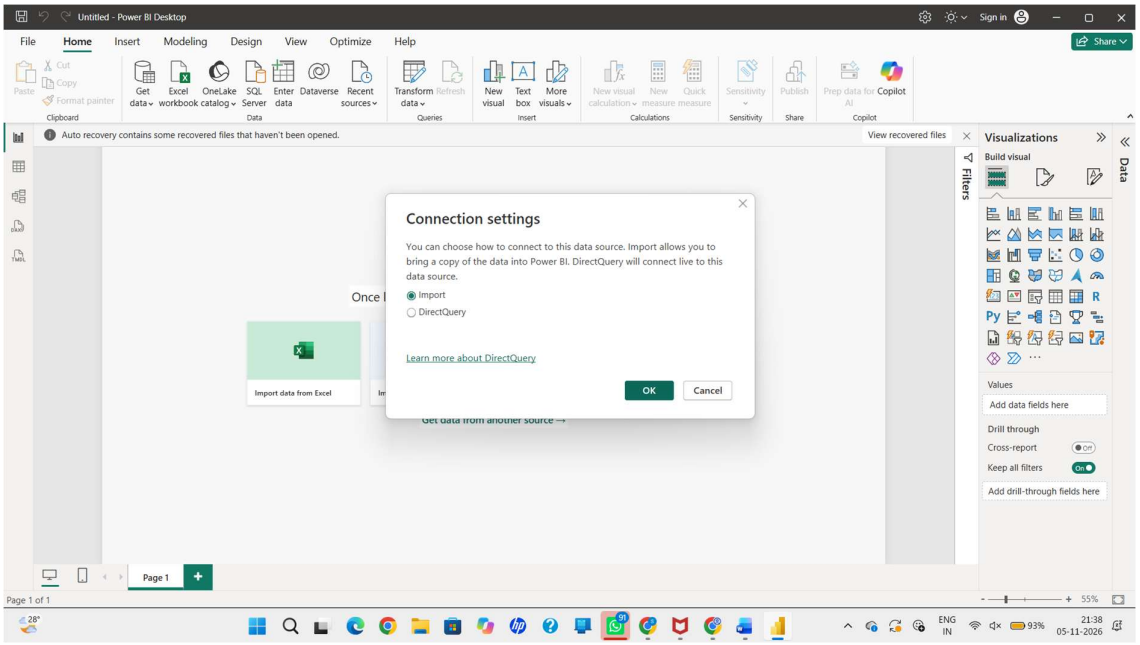
Provide Server URL and Warehouse Name:



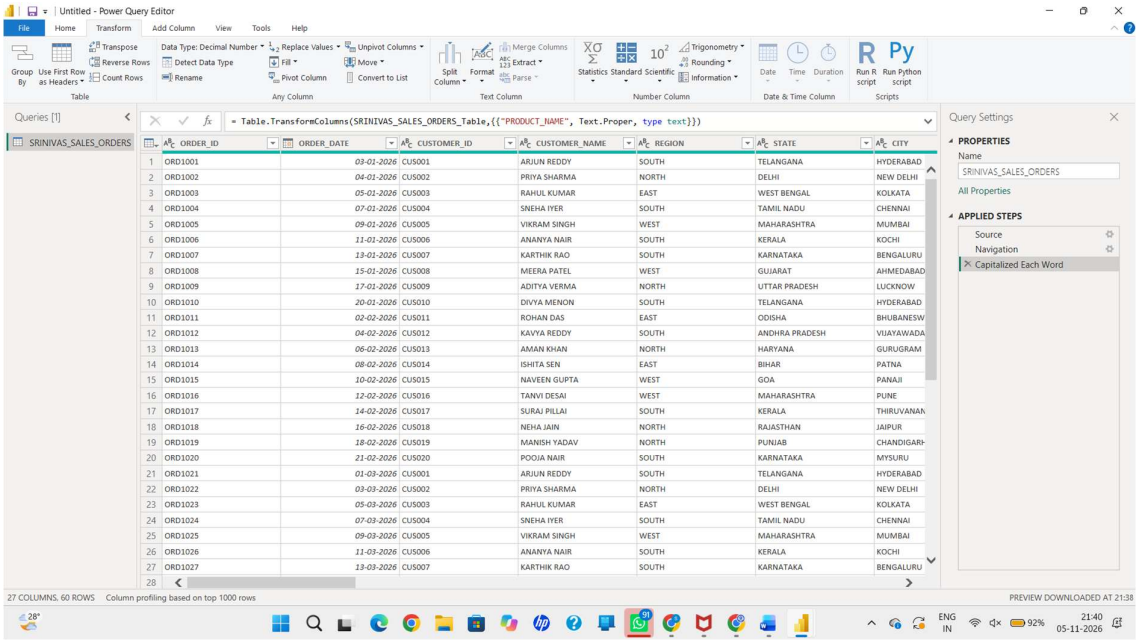
After Providing Username and Password:



Transforming and selecting Import option:



Transform and load the data:



KPI Cards

These cards provide a quick overview of the overall business performance.

1. Total Orders

- Objective: To identify the total number of unique orders placed by customers.
- Explanation: This card displays the distinct count of order IDs for delivered orders. It provides a quick understanding of the number of successfully completed orders.

Power BI title: TOTAL ORDERS

2. Total Revenue

- Objective: To measure the total revenue generated from completed sales.
- Explanation: This card displays the sum of net revenue from delivered orders after applying discounts. It helps understand the overall sales performance.

Power BI title: TOTAL REVENUE

3. Total Customers

- Objective: To identify the total number of unique customers who made purchases.
- Explanation: This card displays the distinct count of customer IDs associated with delivered orders. It helps understand the customer base.

Power BI title: TOTAL CUSTOMERS

4. Average Order Value

- Objective: To calculate the average revenue generated per completed order.

- Explanation: This card displays the average amount spent per order. It helps evaluate the average purchasing value of customers.

Power BI title: AVERAGE ORDER VALUE



5. Revenue by City

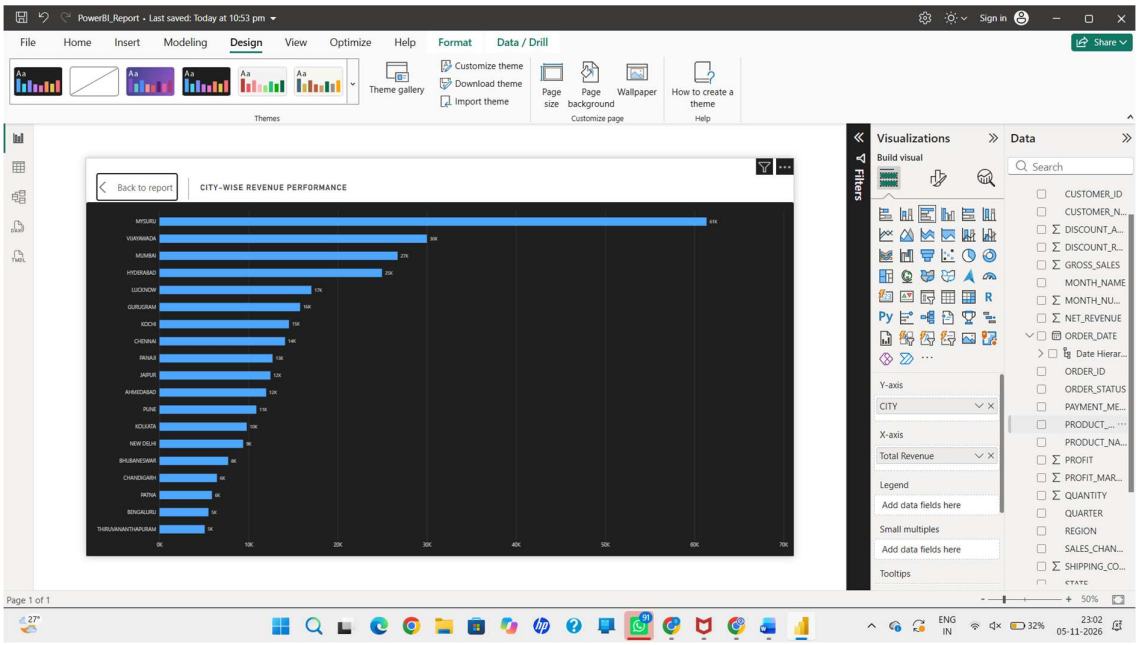
Bar Chart

- Objective: To identify the cities generating the highest revenue.
- Explanation: This chart compares total revenue across different cities. It helps identify high-performing cities and understand the geographical distribution of sales.

Power BI title: CITY-WISE REVENUE PERFORMANCE

Fields:

- Y-axis: CITY
- X-axis: Total Revenue



6. Monthly Revenue Trend

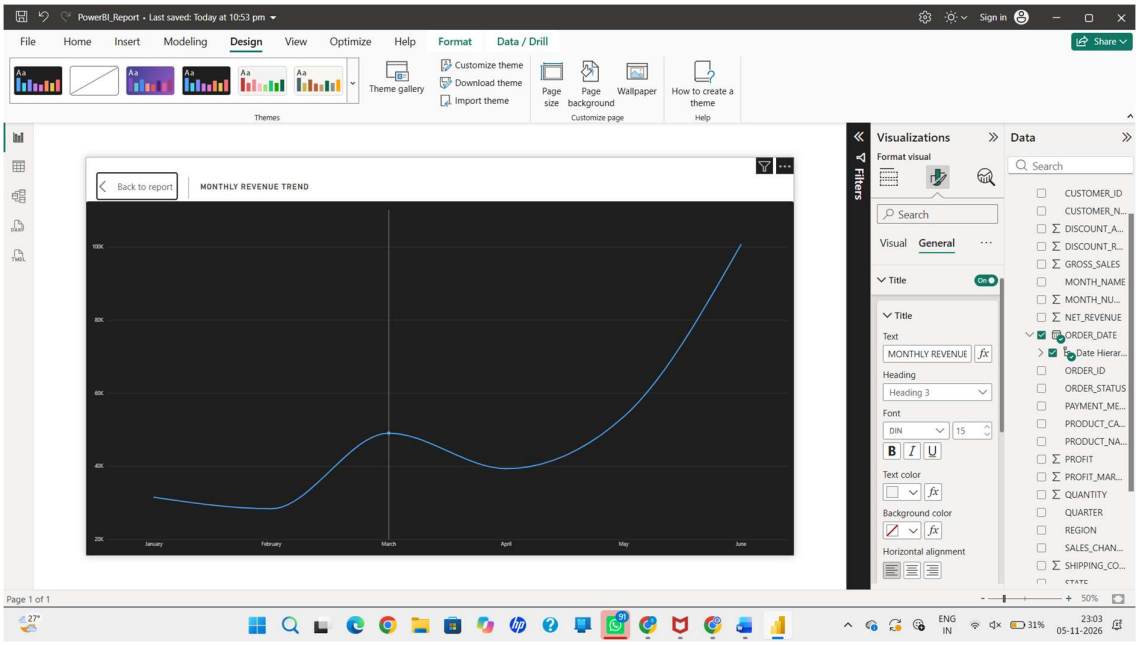
Line Chart

- Objective: To analyze how revenue changes over different months.
- Explanation: This chart displays monthly revenue using the order date hierarchy. It helps identify increasing or decreasing sales trends and understand monthly business performance.

Power BI title: MONTHLY REVENUE TREND

Fields:

- X-axis: ORDER_DATE → Month
- Y-axis: Total Revenue



7. Revenue Distribution by Product Category

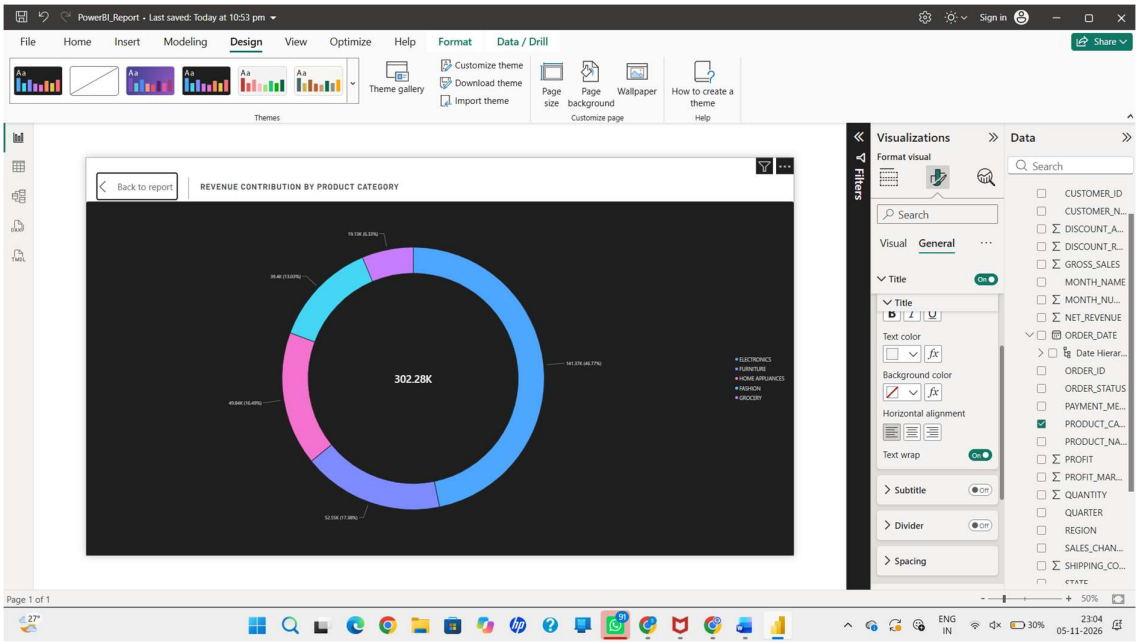
Donut Chart

- Objective: To understand the contribution of different product categories to total revenue.
- Explanation: This chart displays the percentage distribution of revenue across product categories. It helps identify which categories contribute the most to overall sales.

Power BI title: REVENUE CONTRIBUTION BY PRODUCT CATEGORY

Fields:

- Legend: PRODUCT_CATEGORY
- Values: Total Revenue



Additional Analytical Charts

8. Profit by Product Category

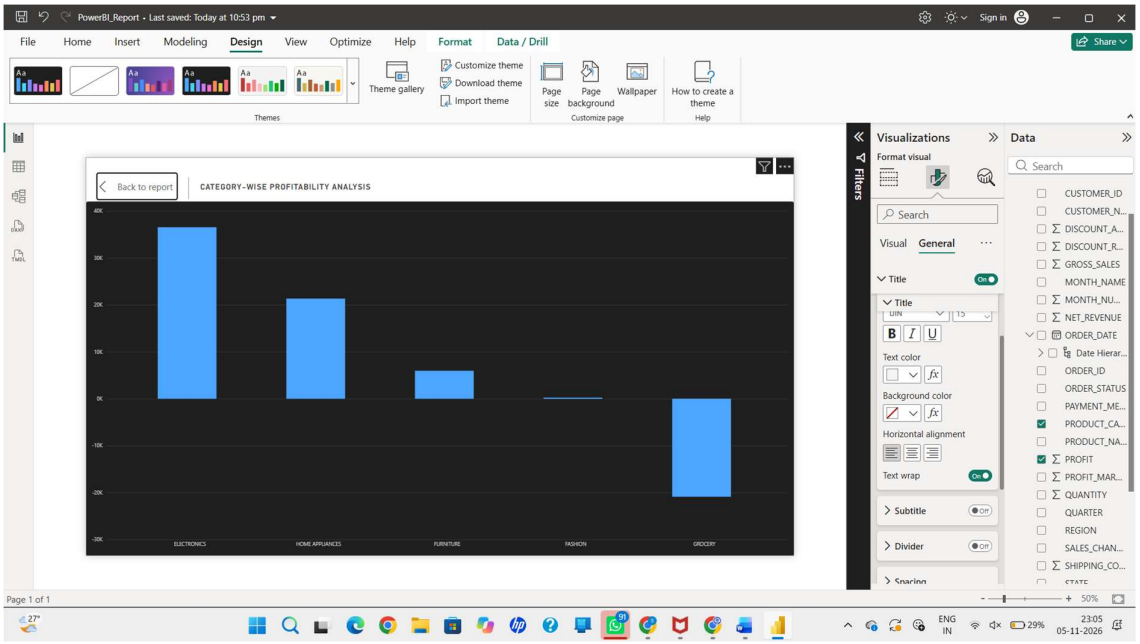
Clustered Column Chart

- Objective: To identify which product categories generate the highest profit.
- Explanation: This chart compares the total profit generated by each product category. It helps the business identify profitable categories and understand their contribution to overall profitability.

Power BI title: CATEGORY-WISE PROFITABILITY ANALYSIS

Fields:

- X-axis: PRODUCT_CATEGORY
- Y-axis: Total Profit



9. Top 6 Customers by Revenue

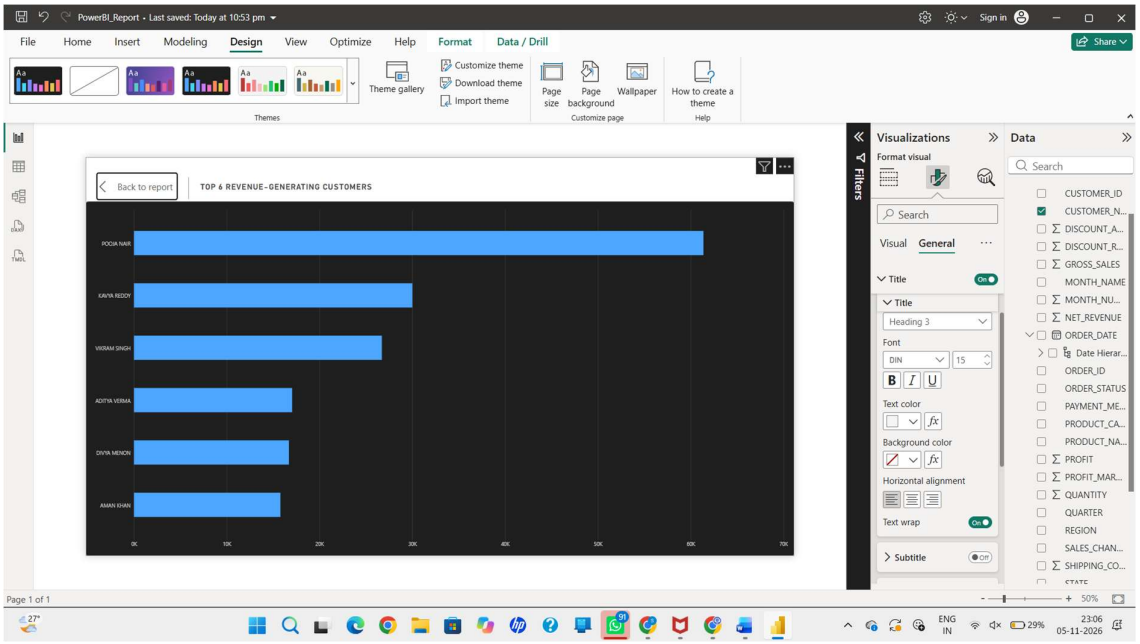
Bar Chart

- Objective: To identify the customers contributing the highest revenue.
- Explanation: This chart displays the top ten customers based on their total purchase revenue. It helps identify valuable customers and understand customer contribution to business performance.

Power BI title: TOP 6 REVENUE-GENERATING CUSTOMERS

Fields:

- Y-axis: CUSTOMER_NAME
- X-axis: Total Revenue
- Filter: Top 6 customers



10. Order Distribution by Region

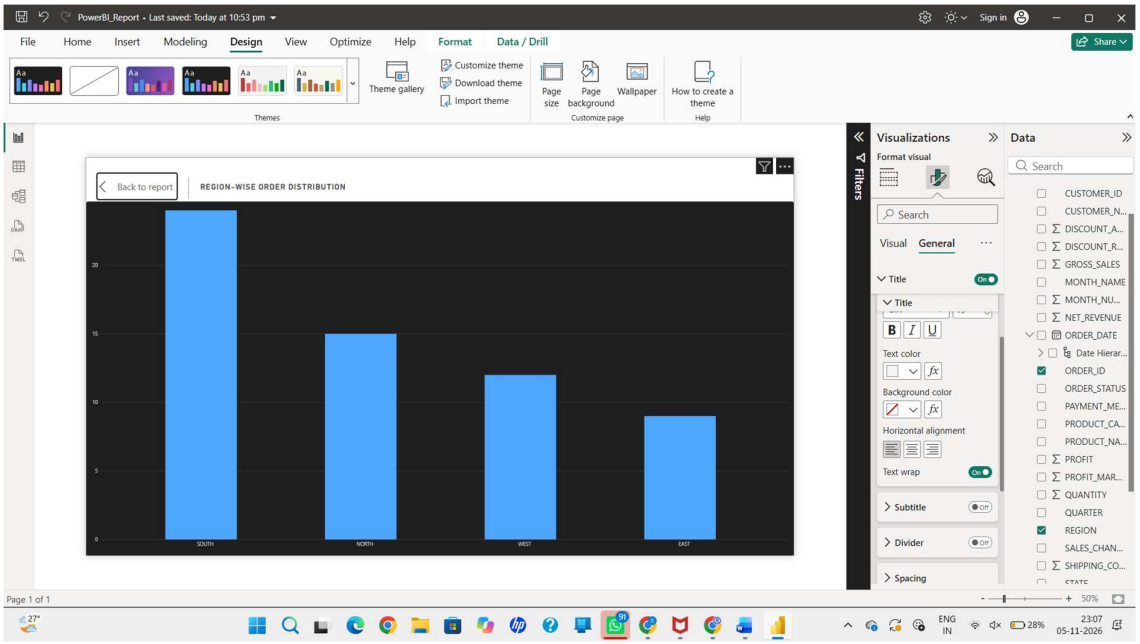
Column Chart

- Objective: To compare the number of orders received from different regions.
- Explanation: This chart displays the number of unique orders across different regions. It helps identify regions with higher order volumes and understand geographical demand.

Power BI title: REGION-WISE ORDER DISTRIBUTION

Fields:

- X-axis: REGION
- Y-axis: Total Orders



11. Revenue Distribution by Product

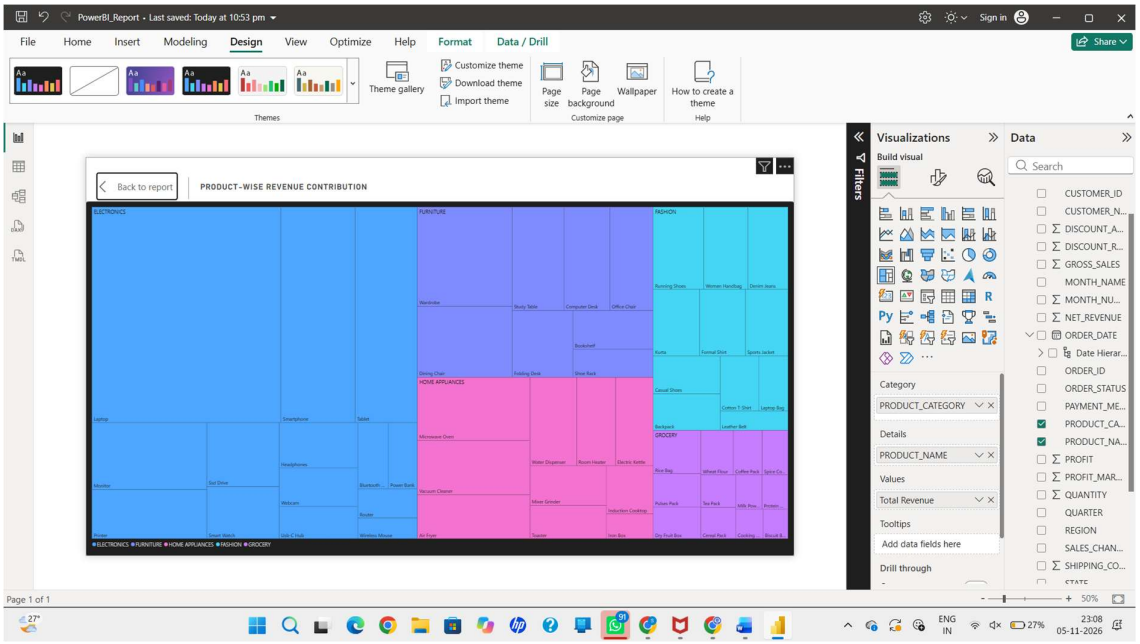
Treemap

- Objective: To identify individual products contributing the most revenue.
- Explanation: This visualization displays products as differently sized rectangles based on their revenue contribution. Larger rectangles represent products generating higher revenue. It helps identify the best-performing products.

Power BI title: PRODUCT-WISE REVENUE CONTRIBUTION

Fields:

- Group: PRODUCT_CATEGORY
- Details: PRODUCT_NAME
- Values: Total Revenue



12. Revenue vs Profit Analysis

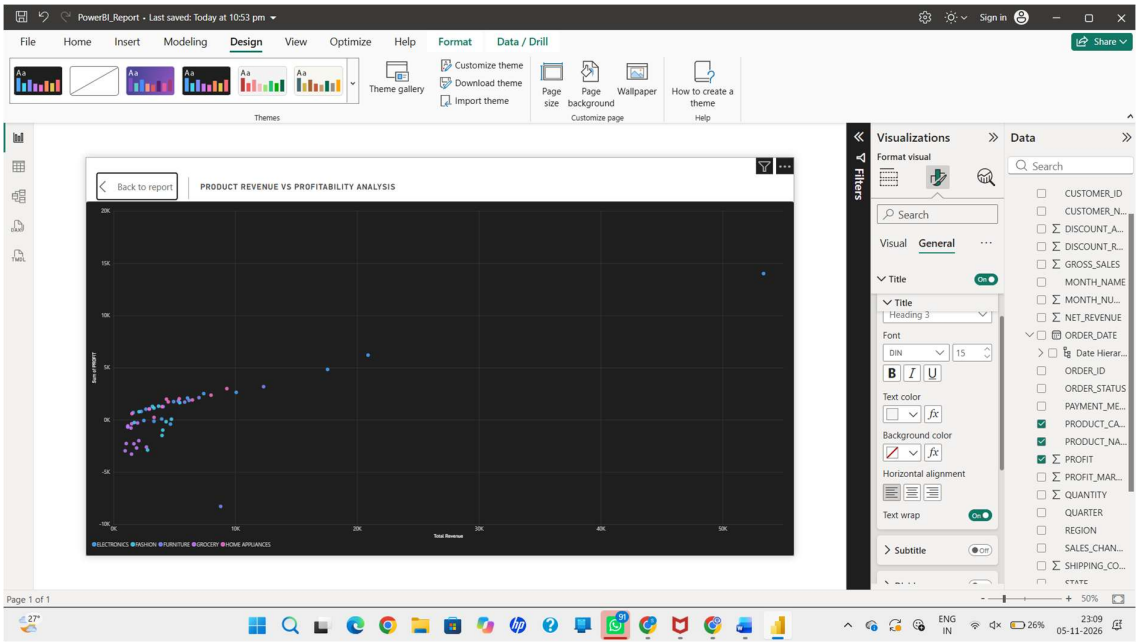
Scatter Chart

- Objective: To understand the relationship between product revenue and profitability.
- Explanation: This chart compares revenue and profit for individual products. It helps identify products that generate high revenue and high profit, as well as products with high revenue but relatively low profitability.

Power BI title: PRODUCT REVENUE VS PROFITABILITY ANALYSIS

Fields:

- X-axis: Total Revenue
- Y-axis: Total Profit
- Details: PRODUCT_NAME
- Legend: PRODUCT_CATEGORY



13. Order Trends Using Date Hierarchy

Line Chart

- Objective: To analyze order volumes across different time periods.
- Explanation: This chart uses the date hierarchy to analyze order counts at different levels, including year, quarter, month, and day. It allows users to drill down into specific periods and understand changes in order activity.

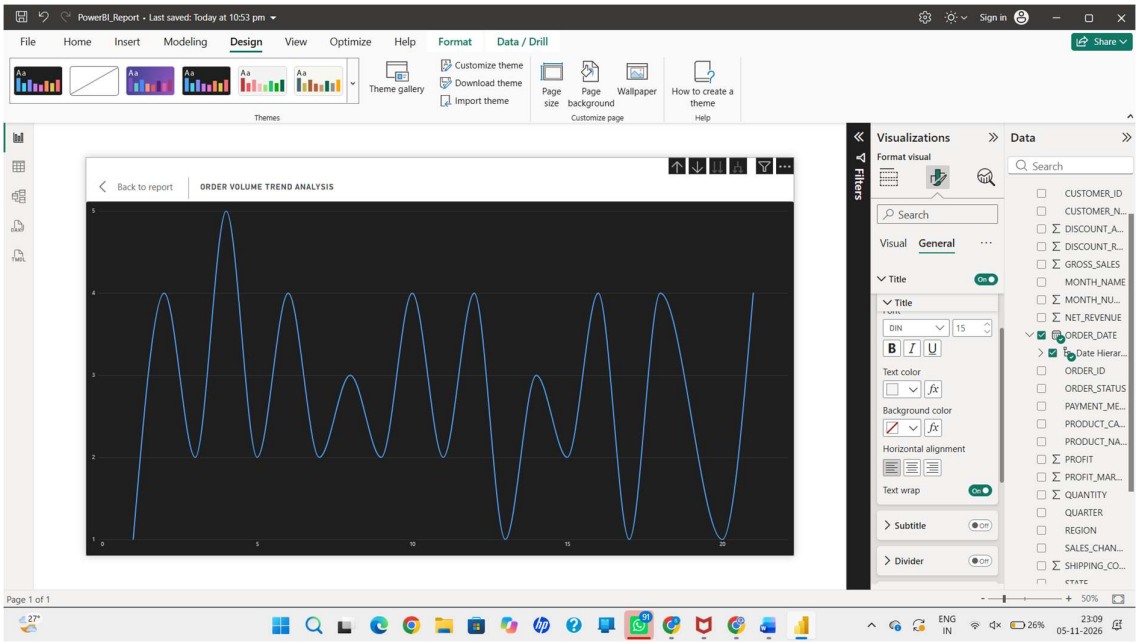
Power BI title: ORDER VOLUME TREND ANALYSIS

Fields:

- X-axis: ORDER_DATE hierarchy
- Y-axis: Total Orders

Hierarchy levels:

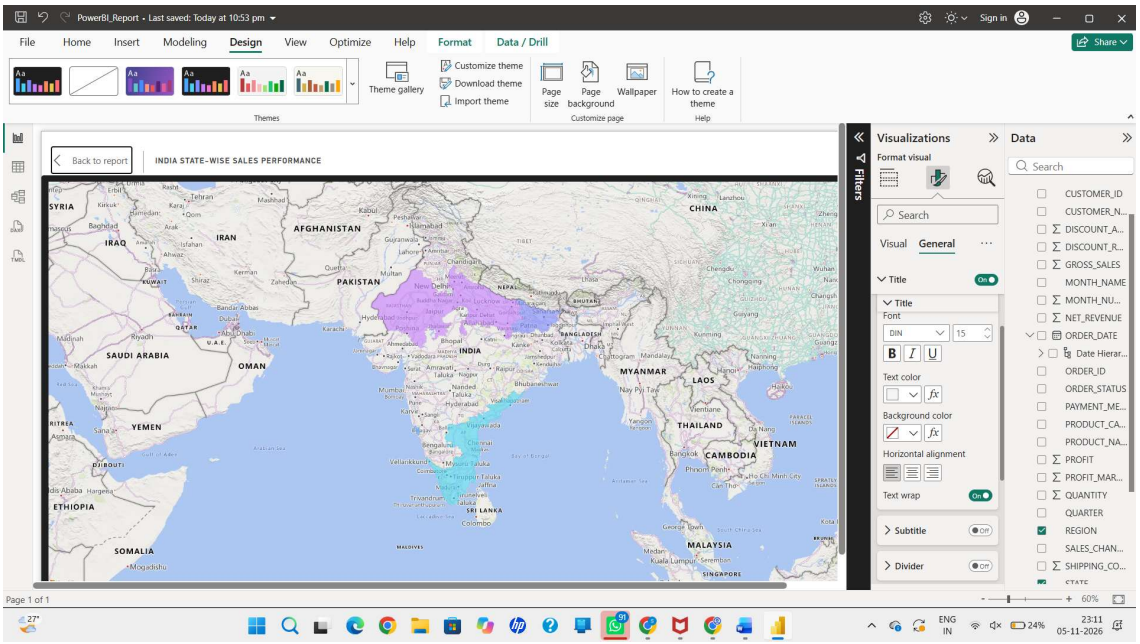
1. Year
2. Quarter
3. Month
4. Day



14. State-Wise Sales Distribution – India Map

- Objective: To analyze the geographical distribution of sales across Indian states.
- Explanation: This map displays Indian states based on their sales performance. Different color intensities can represent the order volume or revenue generated by each state. It helps identify high-performing states and geographical sales patterns.

Power BI title: INDIA STATE-WISE SALES PERFORMANCE



Slicers

Slicers help users interact with the dashboard and filter all relevant visualizations.

15. Product Category Filter

- Objective: To filter sales information based on product categories.
- Explanation: This slicer allows users to select a particular product category and view its revenue, profit, customer contribution and other related information.

Power BI title: FILTER BY PRODUCT CATEGORY



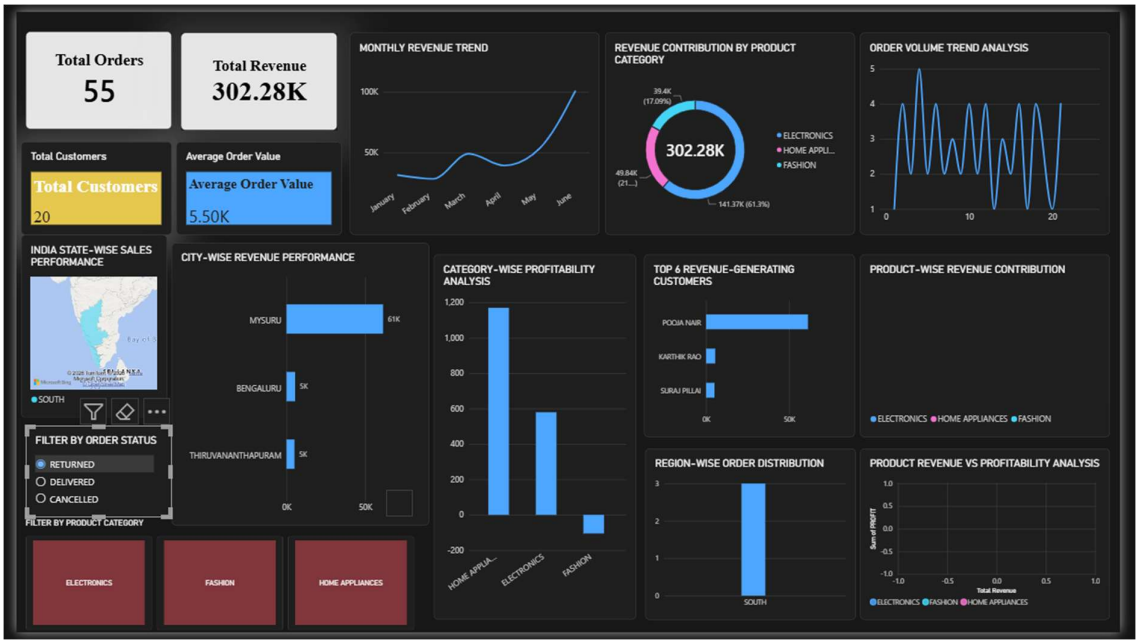
Slicer by Home Appliances



16. Order Status Filter

- Objective: To analyze orders based on their current status.
- Explanation: This slicer allows users to filter delivered, returned and cancelled orders to understand order completion and return patterns.

Power BI title: FILTER BY ORDER STATUS



FINAL REPORT

